

Partial Result: Two c_0 Vector-Measure Range Subcases

Automatic Math Discovery Run `fa.banach.001`

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Source Problem

The source is Jose Rodriguez, *Open problems in Banach spaces and measure theory*. The arXiv identifier is 1607.07787.

Problem 4.2 appears on PDF page 12 and is attributed there to Anantharaman and Diestel:

Let K be a weakly compact subset of c_0 . Is K contained in the range of a c_0 -valued vector measure?

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Problem 4.2 (Anantharaman-Diestel, [4]). *Let K be a weakly compact subset of c_0 . Is K contained in the range of a c_0 -valued vector measure?*

Certain properties of a vector measure are determined by its range. For instance, if ν_1 and ν_2 are vector measures such that $\mathcal{R}(\nu_1) = \mathcal{R}(\nu_2)$, then $|\nu_1|$ is finite if and only if $|\nu_2|$ is finite, [109]. In such case, ν_1 is Bochner differentiable with respect to $|\nu_1|$ if and only if ν_2 is Bochner differentiable with respect to $|\nu_2|$, [110]. An example of Fremlin and Talagrand [61] was used in [111] to show that the previous result is not true in general for the Pettis integral

Definitions

For a countably additive vector measure $\nu : \Sigma \rightarrow X$, its range is

$$\mathcal{R}(\nu) = \{\nu(A) : A \in \Sigma\}.$$

Here countable additivity is meant in the norm of the target Banach space. The problem asks whether every weakly compact $K \subseteq c_0$ is contained in $\mathcal{R}(\nu)$ for some c_0 -valued vector measure ν .

For $a = (a_n) \in c_0$ with $a_n \geq 0$, write

$$Q(a) = \{x \in c_0 : |x_n| \leq a_n \text{ for every } n\}.$$

Idea of the Proof

The coordinate-box case is realized one coordinate at a time. Put two atomless probability intervals behind each coordinate and let the n -th coordinate of the vector measure record a signed amount of mass from those two intervals, scaled by a_n . The tail condition $a_n \rightarrow 0$ supplies the uniform c_0 -control needed for countable additivity.

The second construction shows that the first theorem is not merely a compact set observation. The set $\{0, e_1, e_2, \dots\}$ is weakly compact in c_0 but not norm compact. Rademacher functions give a vector measure whose value on the half-space $\{r_n = 1\}$ is exactly e_n .

Theorem 1 (coordinate boxes). *Let $a = (a_n) \in c_0$ with $a_n \geq 0$. Then $Q(a)$ is contained in the range of a countably additive c_0 -valued vector measure. Consequently, every norm-compact subset of c_0 is contained in such a range.*

Proof. Let

$$\Omega = \bigsqcup_{n \geq 1} (I_n^+ \sqcup I_n^-),$$

where each I_n^+ and I_n^- is a copy of $[0, 1]$ with Lebesgue measure. Let Σ be the sigma algebra obtained from the disjoint union. For $A \in \Sigma$, define

$$\nu(A)_n = a_n (\lambda(A \cap I_n^+) - \lambda(A \cap I_n^-)).$$

Then $|\nu(A)_n| \leq a_n$, so $\nu(A) \in c_0$.

We check countable additivity in c_0 . Let $A_1, A_2, \dots$ be pairwise disjoint and put $B_m = \bigcup_{j > m} A_j$. Since each coordinate is a finite scalar signed measure, for every fixed n we have $\nu(B_m)_n \rightarrow 0$. Given $\varepsilon > 0$, choose N such that $\sup_{n > N} a_n < \varepsilon$. For the finitely many coordinates $n \leq N$, choose m so large that $|\nu(B_m)_n| < \varepsilon$ for all $n \leq N$. Then $\|\nu(B_m)\|_\infty \leq \varepsilon$. Hence the finite partial sums converge in c_0 to $\nu(\bigcup_j A_j)$.

Now fix $x \in Q(a)$. For each n with $a_n > 0$, choose a measurable set $B_n \subseteq I_n^+$ of measure x_n/a_n if $x_n \geq 0$, and choose $B_n \subseteq I_n^-$ of measure $-x_n/a_n$ if $x_n < 0$. If $a_n = 0$, then necessarily $x_n = 0$, and we take $B_n = \emptyset$. With $B = \bigcup_n B_n$, the definition gives $\nu(B) = x$. Thus $Q(a) \subseteq \mathcal{R}(\nu)$.

Finally, if $K \subseteq c_0$ is norm compact, set

$$a_n = \sup_{x \in K} |x_n|.$$

Compactness of K implies $a_n \rightarrow 0$: otherwise a finite epsilon-net in c_0 would give uniformly small tails for all points of K , a contradiction. Therefore $K \subseteq Q(a)$, and the first part applies. $\square$

Theorem 2 (the canonical weakly compact sequence). *The set $\{0\} \cup \{e_n : n \in \mathbb{N}\} \subseteq c_0$ is contained in the range of a countably additive c_0 -valued vector measure.*

Proof. Let $(r_n)_{n \geq 1}$ be the standard Rademacher functions on $[0, 1]$. For Lebesgue measurable $A \subseteq [0, 1]$, define

$$\nu(A) = \left(2 \int_A r_n(t) dt \right)_{n \geq 1}.$$

We first show that $\nu(A) \in c_0$. Approximate 1_A in $L^1[0, 1]$ by a dyadic step function s with mesh 2^{-m} . For every $n > m$, $\int_0^1 s(t) r_n(t) dt = 0$. Hence

$$\left| \int_A r_n(t) dt \right| = \left| \int_0^1 (1_A(t) - s(t)) r_n(t) dt \right| \leq \|1_A - s\|_1$$

for all $n > m$. Since the approximation error can be arbitrarily small, $\int_A r_n \rightarrow 0$, so $\nu(A) \in c_0$.

Countable additivity is also in the norm of c_0 . If $(A_j)_{j \geq 1}$ are pairwise disjoint and $C_m = \bigcup_{j > m} A_j$, then $\lambda(C_m) \rightarrow 0$. Thus

$$\|\nu(C_m)\|_\infty \leq 2\lambda(C_m) \rightarrow 0,$$

which is exactly the norm convergence of the tails of the countable sum.

Let $E_n = \{t : r_n(t) = 1\}$. Orthogonality and independence of the Rademacher functions give

$$2 \int_{E_n} r_j(t) dt = \delta_{jn}.$$

Therefore $\nu(E_n) = e_n$, while $\nu(\emptyset) = 0$. This proves that $\{0\} \cup \{e_n : n \in \mathbb{N}\} \subseteq \mathcal{R}(\nu)$. $\square$

Remark 1. *The set $\{0\} \cup \{e_n : n \in \mathbb{N}\}$ is weakly compact in c_0 , since $e_n \rightarrow 0$ weakly against the dual space ℓ_1 . It is not norm compact, because $\|e_n - e_m\|_\infty = 1$ for $n \neq m$. Thus the Rademacher theorem is a genuine non-norm-compact positive instance of the source problem.*

Scope and Limitations

This packet is a partial result only. It proves the Anantharaman-Diestel question for norm-compact subsets of c_0 , and it gives one basic weakly compact non-norm-compact example. It does not settle arbitrary weakly compact subsets of c_0 , where no coordinatewise null majorant need exist and where the weak compactness may encode substantially more complicated geometry than the single weakly null basis sequence.

Novelty and Search Notes

The run indexes were searched for arXiv:1607.07787, for the phrases “Anantharaman-Diestel”, “weakly compact subset of c_0 ”, “vector measure”, “range of a c_0 -valued vector measure”, and for nearby Banach-space measure theory keywords. No existing run packet for Problem 4.2 or for these two subcases was found. The source paper itself presents Problem 4.2 as open.

This result is constructive and elementary. No later-paper identification is being claimed; the packet should be reviewed as a newly generated partial answer to the extracted problem.

Human Review Recommendation

Review the two norm-convergence arguments for countable additivity. In the coordinate-box construction, the only uniformity input is $a_n \rightarrow 0$. In the Rademacher construction, the only uniformity input for additivity is $\lambda(C_m) \rightarrow 0$ for residual unions in a finite measure space. The containment computations are coordinatewise.

References

- [1] J. Rodriguez, *Open problems in Banach spaces and measure theory*, arXiv:1607.07787.
- [2] R. Anantharaman and J. Diestel, The problem cited as the source of Problem 4.2 in Rodriguez, *Open problems in Banach spaces and measure theory*.